

Question 01.

Create database name Airport_Management_System

```
mysql> CREATE DATABASE Airport_management_system;  
Query OK, 1 row affected (0.01 sec)  
  
mysql> use airport_management_system;  
Database changed  
mysql> |
```

Create tables and insert data

- Airline Table

```
mysql> CREATE TABLE Airline (  
->     airline_id INT PRIMARY KEY,  
->     airline_name VARCHAR(100),  
->     country VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> DESC Airline;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| airline_id | int           | NO   | PRI | NULL    |       |  
| airline_name | varchar(100)  | YES  |     | NULL    |       |  
| country     | varchar(50)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.01 sec)  
  
mysql> |
```

Insert data into Airline table

```
mysql> INSERT INTO Airline VALUES
-> (1, 'SkyWings', 'USA'),
-> (2, 'BlueBird Airways', 'UK'),
-> (3, 'Oceanic Air', 'Australia'),
-> (4, 'FalconFly', 'UAE'),
-> (5, 'GlobalJet', 'Germany');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Airline ;
```

airline_id	airline_name	country
1	SkyWings	USA
2	BlueBird Airways	UK
3	Oceanic Air	Australia
4	FalconFly	UAE
5	GlobalJet	Germany

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

- Flight Table

```
mysql> CREATE TABLE Flight (
-> flight_id INT PRIMARY KEY,
-> airline_id INT,
-> flight_number VARCHAR(20),
-> origin VARCHAR(50),
-> destination VARCHAR(50),
-> departure_time DATETIME,
-> arrival_time DATETIME,
-> FOREIGN KEY (airline_id) REFERENCES Airline(airline_id)
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DESC Flight;
```

Field	Type	Null	Key	Default	Extra
flight_id	int	NO	PRI	NULL	
airline_id	int	YES	MUL	NULL	
flight_number	varchar(20)	YES		NULL	
origin	varchar(50)	YES		NULL	
destination	varchar(50)	YES		NULL	
departure_time	datetime	YES		NULL	
arrival_time	datetime	YES		NULL	

```
7 rows in set (0.00 sec)
```

```
mysql> |
```

Insert data into Flight table

```
mysql> INSERT INTO Flight VALUES
-> (1, 1, 'SW101', 'New York', 'Los Angeles', '2025-08-01 08:00:00', '2025-08-01 11:30:00'),
-> (2, 2, 'BB202', 'London', 'Paris', '2025-08-02 09:00:00', '2025-08-02 10:20:00'),
-> (3, 3, 'OA303', 'Sydney', 'Melbourne', '2025-08-03 07:15:00', '2025-08-03 08:45:00'),
-> (4, 4, 'FF404', 'Dubai', 'Mumbai', '2025-08-04 14:00:00', '2025-08-04 17:15:00'),
-> (5, 5, 'GJ505', 'Berlin', 'Rome', '2025-08-05 10:00:00', '2025-08-05 12:10:00');
Query OK, 5 rows affected (0.00 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Flight;
+-----+-----+-----+-----+-----+-----+-----+
| flight_id | airline_id | flight_number | origin | destination | departure_time | arrival_time |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | 1 | SW101 | New York | Los Angeles | 2025-08-01 08:00:00 | 2025-08-01 11:30:00 |
| 2 | 2 | BB202 | London | Paris | 2025-08-02 09:00:00 | 2025-08-02 10:20:00 |
| 3 | 3 | OA303 | Sydney | Melbourne | 2025-08-03 07:15:00 | 2025-08-03 08:45:00 |
| 4 | 4 | FF404 | Dubai | Mumbai | 2025-08-04 14:00:00 | 2025-08-04 17:15:00 |
| 5 | 5 | GJ505 | Berlin | Rome | 2025-08-05 10:00:00 | 2025-08-05 12:10:00 |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

- Ticket Table

```
mysql> CREATE TABLE Ticket (
->     ticket_id INT PRIMARY KEY,
->     flight_id INT,
->     ticket_price DECIMAL(10,2),
->     ticket_class VARCHAR(20),
->     FOREIGN KEY (flight_id) REFERENCES Flight(flight_id)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> DESC Ticket;
```

Field	Type	Null	Key	Default	Extra
ticket_id	int	NO	PRI	NULL	
flight_id	int	YES	MUL	NULL	
ticket_price	decimal(10,2)	YES		NULL	
ticket_class	varchar(20)	YES		NULL	

4 rows in set (0.00 sec)

```
mysql> |
```

Insert data into Ticket table

```
mysql> INSERT INTO Ticket VALUES
-> (1, 1, 199.99, 'Economy'),
-> (2, 2, 89.50, 'Business'),
-> (3, 3, 120.75, 'Economy'),
-> (4, 4, 220.00, 'First'),
-> (5, 5, 150.00, 'Economy');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Ticket ;
+-----+-----+-----+-----+
| ticket_id | flight_id | ticket_price | ticket_class |
+-----+-----+-----+-----+
| 1 | 1 | 199.99 | Economy |
| 2 | 2 | 89.50 | Business |
| 3 | 3 | 120.75 | Economy |
| 4 | 4 | 220.00 | First |
| 5 | 5 | 150.00 | Economy |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

- Passenger Table

```
mysql> CREATE TABLE Passenger (
->     passenger_id INT PRIMARY KEY,
->     first_name VARCHAR(50),
->     last_name VARCHAR(50),
->     email VARCHAR(100),
->     age INT,
->     seat_number VARCHAR(10),
->     ticket_id INT,
->     FOREIGN KEY (ticket_id) REFERENCES Ticket(ticket_id)
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> DESC Passenger;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| passenger_id | int | NO | PRI | NULL | |
| first_name | varchar(50) | YES | | NULL | |
| last_name | varchar(50) | YES | | NULL | |
| email | varchar(100) | YES | | NULL | |
| age | int | YES | | NULL | |
| seat_number | varchar(10) | YES | | NULL | |
| ticket_id | int | YES | MUL | NULL | |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> |
```

Insert data into Passenger Table

```
mysql> INSERT INTO Passenger VALUES
-> (1, 'John', 'Doe', 'john.doe@example.com', 30, '12A', 1),
-> (2, 'Emily', 'Smith', 'emily.smith@example.com', 27, '4B', 2),
-> (3, 'Carlos', 'Garcia', 'carlos.garcia@example.com', 35, '8C', 3),
-> (4, 'Aisha', 'Khan', 'aisha.khan@example.com', 22, '1A', 4),
-> (5, 'Luca', 'Rossi', 'luca.rossi@example.com', 40, '16D', 5);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Passenger ;
```

passenger_id	first_name	last_name	email	age	seat_number	ticket_id
1	John	Doe	john.doe@example.com	30	12A	1
2	Emily	Smith	emily.smith@example.com	27	4B	2
3	Carlos	Garcia	carlos.garcia@example.com	35	8C	3
4	Aisha	Khan	aisha.khan@example.com	22	1A	4
5	Luca	Rossi	luca.rossi@example.com	40	16D	5

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

Sub Questions of Q01

- Add new column rating to airline table

```
mysql> use airport_management_system;
Database changed
mysql> ALTER TABLE Airline
-> ADD rating DECIMAL(3,1);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC airline;
```

Field	Type	Null	Key	Default	Extra
airline_id	int	NO	PRI	NULL	
airline_name	varchar(100)	YES		NULL	
country	varchar(50)	YES		NULL	
rating	decimal(3,1)	YES		NULL	

```
4 rows in set (0.02 sec)
```

```
mysql> |
```

- ii. Rename column seat_number to seat_code from Passenger Table

```
mysql> ALTER TABLE Passenger
-> RENAME COLUMN seat_number TO seat_code;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC Passenger;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| passenger_id   | int           | NO   | PRI | NULL    |       |
| first_name     | varchar(50)   | YES  |     | NULL    |       |
| last_name      | varchar(50)   | YES  |     | NULL    |       |
| email          | varchar(100)  | YES  |     | NULL    |       |
| age            | int           | YES  |     | NULL    |       |
| seat_code      | varchar(10)   | YES  |     | NULL    |       |
| ticket_id      | int           | YES  | MUL | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql>
```

- iii. Modify ticket_class data type value from Ticket table

```
mysql> ALTER TABLE Ticket
-> MODIFY ticket_class VARCHAR(50);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC Ticket;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| ticket_id      | int           | NO   | PRI | NULL    |       |
| flight_id      | int           | YES  | MUL | NULL    |       |
| ticket_price    | decimal(10,2) | YES  |     | NULL    |       |
| ticket_class    | varchar(50)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

iv. Drop age column from Passenger Table

```
mysql> ALTER TABLE Passenger
-> DROP COLUMN age;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC Passenger;
```

Field	Type	Null	Key	Default	Extra
passenger_id	int	NO	PRI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
email	varchar(100)	YES		NULL	
seat_code	varchar(10)	YES		NULL	
ticket_id	int	YES	MUL	NULL	

6 rows in set (0.00 sec)

```
mysql> |
```

v. Add new column status to the Flight Table

```
mysql> ALTER TABLE Flight
-> ADD status VARCHAR(20);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESC Flight;
```

Field	Type	Null	Key	Default	Extra
flight_id	int	NO	PRI	NULL	
airline_id	int	YES	MUL	NULL	
flight_number	varchar(20)	YES		NULL	
origin	varchar(50)	YES		NULL	
destination	varchar(50)	YES		NULL	
departure_time	datetime	YES		NULL	
arrival_time	datetime	YES		NULL	
status	varchar(20)	YES		NULL	

8 rows in set (0.00 sec)

```
mysql> |
```

Question 02.

Create Database name Bookshop

```
mysql> CREATE DATABASE Bookshop;  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> use Bookshop;  
Database changed
```

I. Create tables and insert data into it

Book Table

```
mysql> use Bookshop;  
Database changed  
mysql> CREATE TABLE Book (  
-> ISBN VARCHAR(20) PRIMARY KEY,  
-> Name VARCHAR(100),  
-> Price DECIMAL(10,2),  
-> Publish_Date DATE  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DESC book;
```

Field	Type	Null	Key	Default	Extra
ISBN	varchar(20)	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Publish_Date	date	YES		NULL	

```
4 rows in set (0.01 sec)
```

```
mysql> |
```

Insert data into book table

```
mysql> INSERT INTO Book VALUES
-> ('ISBN001', 'Madol Doova', 1200.00, '1947-01-01'),
-> ('ISBN002', 'Gamperaliya', 1500.00, '1944-01-01'),
-> ('ISBN003', 'Viragaya', 950.00, '1956-01-01'),
-> ('ISBN004', 'Kalyana Mitra', 800.00, '2010-05-12'),
-> ('ISBN005', 'Harry Potter', 3500.00, '1997-06-26'),
-> ('ISBN006', 'The Hobbit', 2800.00, '1937-09-21'),
-> ('ISBN007', 'Sherlock Holmes', 2200.00, '1892-10-14'),
-> ('ISBN008', 'Senkottan', 1100.00, '2012-08-10');
```

Query OK, 8 rows affected (0.01 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Book;
```

ISBN	Name	Price	Publish_Date
ISBN001	Madol Doova	1200.00	1947-01-01
ISBN002	Gamperaliya	1500.00	1944-01-01
ISBN003	Viragaya	950.00	1956-01-01
ISBN004	Kalyana Mitra	800.00	2010-05-12
ISBN005	Harry Potter	3500.00	1997-06-26
ISBN006	The Hobbit	2800.00	1937-09-21
ISBN007	Sherlock Holmes	2200.00	1892-10-14
ISBN008	Senkottan	1100.00	2012-08-10

8 rows in set (0.00 sec)

```
mysql> |
```

Author table

```
mysql> CREATE TABLE Author (
->   AuthorID VARCHAR(20) PRIMARY KEY,
->   Name VARCHAR(100),
->   Biography TEXT
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> DESC Author;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| AuthorID   | varchar(20)   | NO   | PRI | NULL    |       |
| Name       | varchar(100)  | YES  |     | NULL    |       |
| Biography  | text          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

Insert data into Author table

```
mysql> INSERT INTO Author VALUES
-> ('A001', 'Martin Wickramasinghe', 'Famous Sri Lankan novelist'),
-> ('A002', 'K. Jayatilleke', 'Prominent Sinhala writer'),
-> ('A003', 'J.K. Rowling', 'British author of Harry Potter'),
-> ('A004', 'J.R.R. Tolkien', 'English writer of high fantasy'),
-> ('A005', 'Arthur Conan Doyle', 'Creator of Sherlock Holmes'),
-> ('A006', 'Mahagama Sekara', 'Famous poet and writer'),
-> ('A007', 'Kumaratunga Munidasa', 'Linguist and grammarian'),
-> ('A008', 'T.B. Ilangaratne', 'Author of children literature');
Query OK, 8 rows affected (0.00 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> SELECT
-> \c
mysql> SELECT * FROM Author;
+-----+-----+-----+
| AuthorID | Name                | Biography                |
+-----+-----+-----+
| A001     | Martin Wickramasinghe | Famous Sri Lankan novelist |
| A002     | K. Jayatilleke       | Prominent Sinhala writer  |
| A003     | J.K. Rowling         | British author of Harry Potter |
| A004     | J.R.R. Tolkien       | English writer of high fantasy |
| A005     | Arthur Conan Doyle   | Creator of Sherlock Holmes |
| A006     | Mahagama Sekara      | Famous poet and writer    |
| A007     | Kumaratunga Munidasa | Linguist and grammarian    |
| A008     | T.B. Ilangaratne     | Author of children literature |
+-----+-----+-----+
8 rows in set (0.00 sec)

mysql> |
```

Customer Table

```
mysql> CREATE TABLE Customer (  
-> CustomerID VARCHAR(20) PRIMARY KEY,  
-> Name VARCHAR(100),  
-> Email VARCHAR(100),  
-> Address VARCHAR(250)  
-> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DESC Customer;
```

Field	Type	Null	Key	Default	Extra
CustomerID	varchar(20)	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Email	varchar(100)	YES		NULL	
Address	varchar(250)	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> |
```

Insert data into customer table

```
mysql> INSERT INTO Customer VALUES  
-> ('C001', 'Ravindu Bandara', 'ravindu.b@gmail.com', 'No. 14/2, Temple Road, Kiribathgoda'),  
-> ('C002', 'Tharushi Perera', 'tharushi.p@outlook.com', 'No. 88, Lake Drive, Rajagiriya'),  
-> ('C003', 'Mahesh Wickramasinghe', 'mahesh.wick@yahoo.com', 'No. 205, Main Street, Anuradhapura'),  
-> ('C004', 'Kavindi Jayawardena', 'kavindi.j@gmail.com', 'No. 42, Flower Road, Colombo 07'),  
-> ('C005', 'Dhanushka Rathnayake', 'dhanushka.r@hotmail.com', 'No. 115, High Level Road, Maharagama'),  
-> ('C006', 'Sachini Alwis', 'sachini.a@gmail.com', 'No. 73, Circular Road, Badulla'),  
-> ('C007', 'Nuwan Senanayake', 'nuwan.sena@gmail.com', 'No. 31/A, Sea Street, Negombo'),  
-> ('C008', 'Shenali Fonseka', 'shenali.f@icloud.com', 'No. 9, Galle Road, Panadura');  
Query OK, 8 rows affected (0.00 sec)  
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM customer;
```

CustomerID	Name	Email	Address
C001	Ravindu Bandara	ravindu.b@gmail.com	No. 14/2, Temple Road, Kiribathgoda
C002	Tharushi Perera	tharushi.p@outlook.com	No. 88, Lake Drive, Rajagiriya
C003	Mahesh Wickramasinghe	mahesh.wick@yahoo.com	No. 205, Main Street, Anuradhapura
C004	Kavindi Jayawardena	kavindi.j@gmail.com	No. 42, Flower Road, Colombo 07
C005	Dhanushka Rathnayake	dhanushka.r@hotmail.com	No. 115, High Level Road, Maharagama
C006	Sachini Alwis	sachini.a@gmail.com	No. 73, Circular Road, Badulla
C007	Nuwan Senanayake	nuwan.sena@gmail.com	No. 31/A, Sea Street, Negombo
C008	Shenali Fonseka	shenali.f@icloud.com	No. 9, Galle Road, Panadura

```
8 rows in set (0.00 sec)
```

```
mysql> |
```

Orders Table

```
mysql> CREATE TABLE Orders (  
  ->   OrderID VARCHAR(20) PRIMARY KEY,  
  ->   CustomerID VARCHAR(20),  
  ->   Order_Date DATE,  
  ->   FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)  
  -> );  
Query OK, 0 rows affected (0.02 sec)  
  
mysql> DESC Orders;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type          | Null | Key  | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| OrderID    | varchar(20)   | NO   | PRI  | NULL    |       |  
| CustomerID | varchar(20)   | YES  | MUL  | NULL    |       |  
| Order_Date | date          | YES  |      | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql> |
```

Insert data into Orders Table

```
mysql> INSERT INTO Orders VALUES  
  -> ('ORD001', 'C001', '2025-02-01'),  
  -> ('ORD002', 'C002', '2025-02-02'),  
  -> ('ORD003', 'C003', '2025-02-03'),  
  -> ('ORD004', 'C004', '2025-02-04'),  
  -> ('ORD005', 'C005', '2025-02-05'),  
  -> ('ORD006', 'C006', '2025-02-06'),  
  -> ('ORD007', 'C007', '2025-02-07'),  
  -> ('ORD008', 'C008', '2025-02-08');  
Query OK, 8 rows affected (0.01 sec)  
Records: 8  Duplicates: 0  Warnings: 0  
  
mysql> SELECT * FROM Orders;  
+-----+-----+-----+  
| OrderID | CustomerID | Order_Date |  
+-----+-----+-----+  
| ORD001  | C001       | 2025-02-01 |  
| ORD002  | C002       | 2025-02-02 |  
| ORD003  | C003       | 2025-02-03 |  
| ORD004  | C004       | 2025-02-04 |  
| ORD005  | C005       | 2025-02-05 |  
| ORD006  | C006       | 2025-02-06 |  
| ORD007  | C007       | 2025-02-07 |  
| ORD008  | C008       | 2025-02-08 |  
+-----+-----+-----+  
8 rows in set (0.00 sec)  
  
mysql> |
```

Review Table

```
mysql> CREATE TABLE Review (  
  -> ReviewID VARCHAR(20) PRIMARY KEY,  
  -> Rating INT,  
  -> Comment TEXT  
  -> );
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DESC Review;
```

Field	Type	Null	Key	Default	Extra
ReviewID	varchar(20)	NO	PRI	NULL	
Rating	int	YES		NULL	
Comment	text	YES		NULL	

3 rows in set (0.00 sec)

```
mysql> |
```

Insert data into Review Table

```
mysql> INSERT INTO Review VALUES
```

```
  -> ('R001', 5, 'Comment 1'),  
  -> ('R002', 4, 'Comment 2'),  
  -> ('R003', 3, 'Comment 3'),  
  -> ('R004', 5, 'Comment 4'),  
  -> ('R005', 2, 'Comment 5'),  
  -> ('R006', 4, 'Comment 6'),  
  -> ('R007', 5, 'Comment 7'),  
  -> ('R008', 1, 'Comment 8');
```

Query OK, 8 rows affected (0.00 sec)

Records: 8 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Review;
```

ReviewID	Rating	Comment
R001	5	Comment 1
R002	4	Comment 2
R003	3	Comment 3
R004	5	Comment 4
R005	2	Comment 5
R006	4	Comment 6
R007	5	Comment 7
R008	1	Comment 8

8 rows in set (0.00 sec)

```
mysql>
```

Q2 sub questions answers

i. I already did it above

II. Add new column(Genre) to book table

```
mysql> ALTER TABLE Book
-> ADD Genre VARCHAR(50);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC Book;
```

Field	Type	Null	Key	Default	Extra
ISBN	varchar(20)	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Publish_Date	date	YES		NULL	
Genre	varchar(50)	YES		NULL	

```
5 rows in set (0.00 sec)

mysql> |
```

III. add new column(Phone) to customer table

```
mysql> ALTER TABLE Customer
-> ADD Phone VARCHAR(15);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC customer;
```

Field	Type	Null	Key	Default	Extra
CustomerID	varchar(20)	NO	PRI	NULL	
Name	varchar(100)	YES		NULL	
Email	varchar(100)	YES		NULL	
Address	varchar(250)	YES		NULL	
Phone	varchar(15)	YES		NULL	

```
5 rows in set (0.00 sec)

mysql> |
```

IV. Rename column order_date to purchaseDate

```
mysql> ALTER TABLE Orders
-> RENAME COLUMN Order_Date TO PurchaseDate;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESC Orders;
```

Field	Type	Null	Key	Default	Extra
OrderID	varchar(20)	NO	PRI	NULL	
CustomerID	varchar(20)	YES	MUL	NULL	
PurchaseDate	date	YES		NULL	

```
3 rows in set (0.00 sec)

mysql> |
```

Question03.

I. Create Database LCport

```
mysql> CREATE DATABASE LCport;
Query OK, 1 row affected (0.00 sec)

mysql> use LCport;
Database changed
```

Create Tables and insert data into tables

II Ship Table

```
mysql> use LCport;
Database changed
mysql> CREATE TABLE Ships(
    -> ship_id INT PRIMARY KEY,
    -> ship_name VARCHAR(200),
    -> origin_port VARCHAR(60),
    -> destination_port VARCHAR(60),
    -> arrival_date DATE,
    -> departure_date DATE
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql> DESC ships;
```

Field	Type	Null	Key	Default	Extra
ship_id	int	NO	PRI	NULL	
ship_name	varchar(200)	YES		NULL	
origin_port	varchar(60)	YES		NULL	
destination_port	varchar(60)	YES		NULL	
arrival_date	date	YES		NULL	
departure_date	date	YES		NULL	

```
6 rows in set (0.00 sec)

mysql> |
```

Insert data into ships table

```
mysql> INSERT INTO ships VALUES
-> (1, "Ocean Pal", "Singapore Port", 'Colombo Port', '2026-03-01', '2026-03-03'),
-> (2, "Sea Express", "Dubai Port", 'Singapore Port', '2026-03-05', '2026-03-07'),
-> (3, "Blue Wave", "Maladiwaina", 'Hamburg', '2026-03-10', '2026-03-12'),
-> (4, "Global Freight", "Shanghai Port", 'Colombo Port', '2026-03-15', '2026-03-18'),
-> (5, "Pacific Star", "Russia", 'Los Angeles', '2026-03-20', '2026-03-23');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM ships;
```

ship_id	ship_name	origin_port	destination_port	arrival_date	departure_date
1	Ocean Pal	Singapore Port	Colombo Port	2026-03-01	2026-03-03
2	Sea Express	Dubai Port	Singapore Port	2026-03-05	2026-03-07
3	Blue Wave	Maladiwaina	Hamburg	2026-03-10	2026-03-12
4	Global Freight	Shanghai Port	Colombo Port	2026-03-15	2026-03-18
5	Pacific Star	Russia	Los Angeles	2026-03-20	2026-03-23

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

III. Containers table

```
mysql> CREATE TABLE Containers (
->   container_id INT PRIMARY KEY,
->   ship_id INT,
->   container_type VARCHAR(50),
->   weight DECIMAL(10,2),
->   status VARCHAR(50),
->   FOREIGN KEY (ship_id) REFERENCES Ships(ship_id)
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> DESC containers;
```

Field	Type	Null	Key	Default	Extra
container_id	int	NO	PRI	NULL	
ship_id	int	YES	MUL	NULL	
container_type	varchar(50)	YES		NULL	
weight	decimal(10,2)	YES		NULL	
status	varchar(50)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

Insert data into Containers table

```
mysql> INSERT INTO Containers VALUES
-> (101, 1, 'Standard Dry', 24500.00, 'Loaded'),
-> (102, 1, 'Refrigerated', 28200.00, 'In Transit'),
-> (103, 2, 'Flat Rack', 32800.00, 'Unloaded'),
-> (104, 3, 'Open Top', 19400.00, 'Loaded'),
-> (105, 4, 'Liquid Tank', 30100.00, 'In Transit');
```

Query OK, 5 rows affected (0.01 sec)

Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM containers;
```

container_id	ship_id	container_type	weight	status
101	1	Standard Dry	24500.00	Loaded
102	1	Refrigerated	28200.00	In Transit
103	2	Flat Rack	32800.00	Unloaded
104	3	Open Top	19400.00	Loaded
105	4	Liquid Tank	30100.00	In Transit

5 rows in set (0.00 sec)

```
mysql> |
```

IV. Cargo table

```
mysql> CREATE TABLE Cargo (
->   cargo_id INT PRIMARY KEY,
->   container_id INT,
->   description VARCHAR(255),
->   weight DECIMAL(10,2),
->   cargo_type VARCHAR(50),
->   FOREIGN KEY (container_id) REFERENCES Containers(container_id)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> DESC Cargo;
```

Field	Type	Null	Key	Default	Extra
cargo_id	int	NO	PRI	NULL	
container_id	int	YES	MUL	NULL	
description	varchar(255)	YES		NULL	
weight	decimal(10,2)	YES		NULL	
cargo_type	varchar(50)	YES		NULL	

5 rows in set (0.00 sec)

```
mysql> |
```

Insert data into Cargo table

```
mysql> INSERT INTO Cargo VALUES
-> (501, 101, 'Electronic Goods', 8.50, 'General'),
-> (502, 102, 'Frozen Seafood', 10.20, 'Perishable'),
-> (503, 103, 'Heavy Machinery', 18.00, 'Oversized'),
-> (504, 104, 'Raw Cotton', 14.30, 'Bulk'),
-> (505, 105, 'Liquid Chemicals', 19.50, 'Hazardous');
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Cargo;
```

cargo_id	container_id	description	weight	cargo_type
501	101	Electronic Goods	8.50	General
502	102	Frozen Seafood	10.20	Perishable
503	103	Heavy Machinery	18.00	Oversized
504	104	Raw Cotton	14.30	Bulk
505	105	Liquid Chemicals	19.50	Hazardous

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

V. Employees table

```
mysql> CREATE TABLE Employees (
->     employee_id INT PRIMARY KEY,
->     full_name VARCHAR(100),
->     role VARCHAR(50),
->     shift VARCHAR(20),
->     contact_number VARCHAR(15)
-> );
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DESC Employees;
```

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
full_name	varchar(100)	YES		NULL	
role	varchar(50)	YES		NULL	
shift	varchar(20)	YES		NULL	
contact_number	varchar(15)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

Insert data into Employees table

```
mysql> INSERT INTO Employees VALUES
-> (1, 'Kasun Perera', 'Crane Operator', 'Morning', '0771546467'),
-> (2, 'Dilini Jayawardena', 'Logistics Officer', 'Day', '0719876879'),
-> (3, 'Tharindu Silva', 'Harbor Master', 'Night', '0753334444'),
-> (4, 'Nipuni Fernando', 'Customs Inspector', 'Morning', '0785556666'),
-> (5, 'Sanjeewa Bandara', 'Safety Officer', 'Night', '0707778888');
Query OK, 5 rows affected (0.00 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Employees;
```

employee_id	full_name	role	shift	contact_number
1	Kasun Perera	Crane Operator	Morning	0771546467
2	Dilini Jayawardena	Logistics Officer	Day	0719876879
3	Tharindu Silva	Harbor Master	Night	0753334444
4	Nipuni Fernando	Customs Inspector	Morning	0785556666
5	Sanjeewa Bandara	Safety Officer	Night	0707778888

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

VI. Schedules table

```
mysql> CREATE TABLE Schedules (
->     schedule_id INT PRIMARY KEY,
->     ship_id INT,
->     berth_number VARCHAR(20),
->     scheduled_arrival DATETIME,
->     scheduled_departure DATETIME,
->     FOREIGN KEY (ship_id) REFERENCES Ships(ship_id)
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DESC schedules;
```

Field	Type	Null	Key	Default	Extra
schedule_id	int	NO	PRI	NULL	
ship_id	int	YES	MUL	NULL	
berth_number	varchar(20)	YES		NULL	
scheduled_arrival	datetime	YES		NULL	
scheduled_departure	datetime	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

Insert data into Schedules table

```
mysql> INSERT INTO Schedules VALUES
  -> (1, 1, 'Berth-A1', '2026-03-01 08:00:00', '2026-03-03 18:00:00'),
  -> (2, 2, 'Berth-B2', '2026-03-05 10:30:00', '2026-03-07 14:00:00'),
  -> (3, 3, 'Berth-C1', '2026-03-10 06:00:00', '2026-03-12 20:00:00'),
  -> (4, 4, 'Berth-A2', '2026-03-15 12:00:00', '2026-03-18 16:30:00'),
  -> (5, 5, 'Berth-D3', '2026-03-20 09:15:00', '2026-03-23 11:00:00');
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM schedules;
```

schedule_id	ship_id	berth_number	scheduled_arrival	scheduled_departure
1	1	Berth-A1	2026-03-01 08:00:00	2026-03-03 18:00:00
2	2	Berth-B2	2026-03-05 10:30:00	2026-03-07 14:00:00
3	3	Berth-C1	2026-03-10 06:00:00	2026-03-12 20:00:00
4	4	Berth-A2	2026-03-15 12:00:00	2026-03-18 16:30:00
5	5	Berth-D3	2026-03-20 09:15:00	2026-03-23 11:00:00

```
5 rows in set (0.00 sec)
```

```
mysql> |
```